

UNIVERSITY OF MINDANAO TAGUM BRANCH
BACHELOR OF SCIENCE IN COMPUTER SCIENCE
Department of Computing Education
Visayan Village, Tagum City, Philippines



Course Requirements
in
Programming Languages (CS 15/L - 1831)

Attendance Management System for School Events

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Team Members

No.	Name	Role
1	Mundiz, Kenn JB A.	Project Manager / Developer
2	San Jose, Alexie Bryce C.	Database / Backend / Developer
3	Buena, Gabriel Jee M.	Frontend / UI/UX / Developer

Project Description

The Attendance Management System is a React Native application designed to help users, specifically students, student officials, and organization members, to manage and supervise student attendance during school events. The application primarily addresses the common problem of having to monitor student attendance during school events manually such as using physical logbooks. With this problem, the application allows users to utilize generated QR code as a means of managing student attendance effectively and efficiently. The application allows students to scan the QR code to automatically record every student attendance into the system.

Target Users

The application primarily targets the following users:

- Students
- Student Officials
- Student Leaders
- Student School Organizations

Objectives

The application primarily aims to:

- Develop a mobile application using React Native that allow users to manage and facilitate student attendance during school events.
- To implement CRUD (Create, Read, Update, Delete) operations using SQLite.
- To provide a straightforward and comprehensible interface for managing student attendance records.

- To implement QR code generation and scanning for efficient attendance recording during school events
- To reduce errors and inconsistencies associated with manual attendance recording.
- To provide real-time recording of student attendance upon successful QR code scanning.
- To enable authorized users to view, search, and monitor attendance records efficiently.

Major Features

- Student role selection
- Generate QR code
- Automatic attendance recording
- View attendance history
- Create event
- Edit event
- Delete event
- View, update, and delete attendees
- Search attendance records
- Scan students' QR code
- SQLite database
- Duplicate attendance prevention
- Automatic date/time recording

Proposed Screens

Screen	Purpose
Login / User selection	Allows users to login in or select their role (Student / Official)
Home	Serves as the user's main dashboard and provides access to QR generation/scanning and attendance history.
Generate QR	Creates QR for attendance.

Scan QR	Scans the QR for attendance
Attendance History/Records	Views the student attendance history/records
Create Event	Officers create school events
Edit Event	Officers edit school events
View Attendees	Officers view the list of student attendees
Edit Attendees	Officers edit the student attendees

Proposed Transaction

Student Officers will enter the necessary information for a school event, such as the event name, date, and time. This information will be stored in the system's SQLite database and used to create and manage the event.

Students will provide their student information and scan the QR code generated for the event. The system will process the scanned information, verify the attendance, and automatically save the student's attendance record along with the date and time. The stored attendance information can then be viewed and searched by authorized users.

Proposed Database

The proposed Attendance Management System will use **SQLite** as its database solution. SQLite is a lightweight relational database that can be integrated into a React Native application through Expo. It will store student information, event details, QR code data, and attendance records directly within the application. SQLite is suitable for the project because it supports CRUD operations and does not require a separate database server.

CRUD Plan

The application will perform the following CRUD operations:

Operation	Proposed Function
CREATE	Create a new school event and add a new attendance record after a student officer scans the QR code
READ	View the list of events, registered attendees, and student attendance history

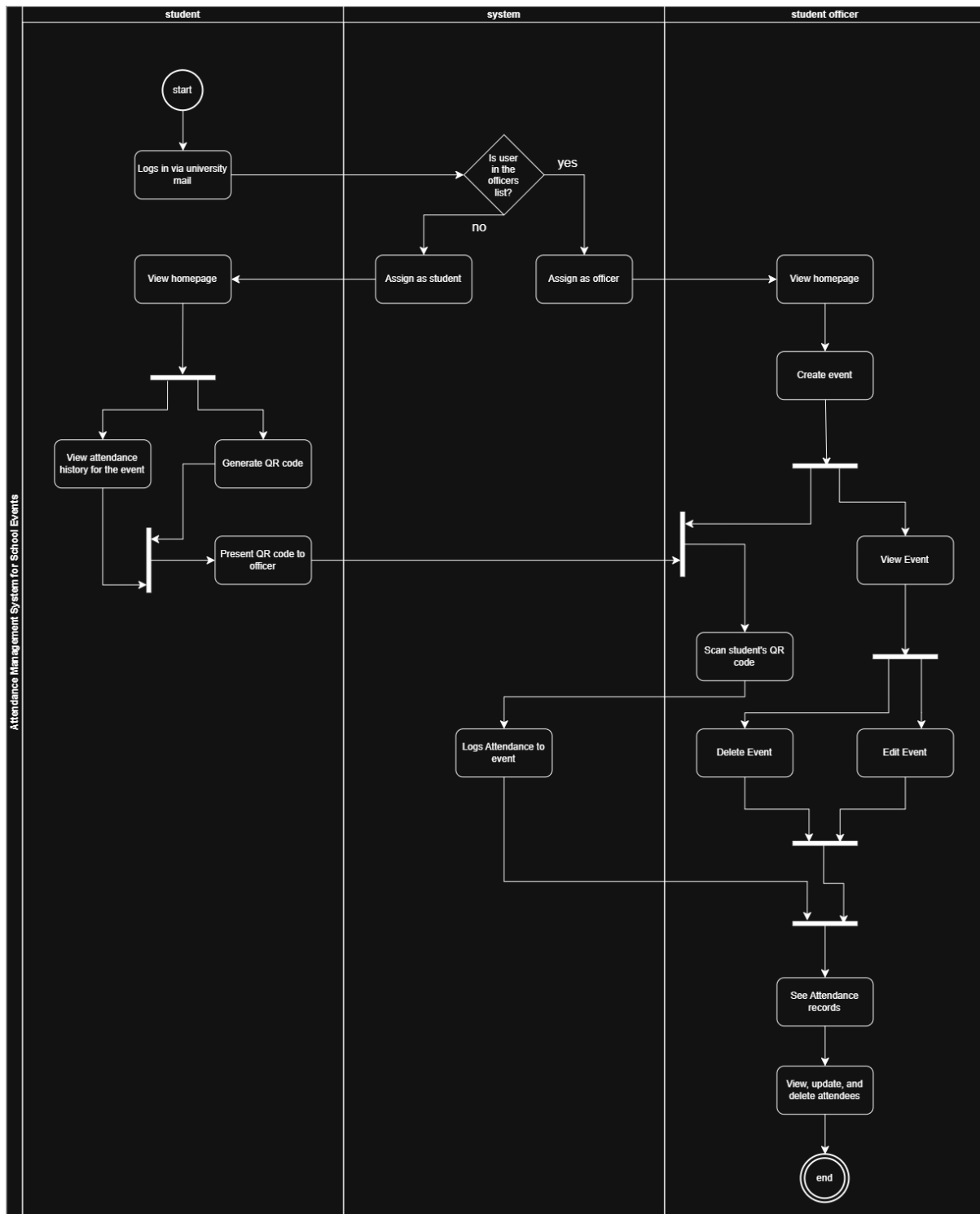
UPDATE	Edit event details and update attendance information
DELETE	Delete an event or remove an incorrect attendance record

Technology Stack

Frontend	React Native
Development Platform	Expo
Programming Language	JavaScript, TypeScript
Navigation	React Navigation
Database	SQLite

Proposed App Architecture / Flow

For this section, we used a UML Activity Diagram with swimlanes to illustrate the flow of the application. The diagram is divided into three swimlanes representing the Student, System, and Student Officer actors. The diagram assumes a single scenario, starting from one initial point and ending at one final point, showing how the actions and information flow between the actors throughout the attendance process.



UI/UX Concept

The Attendance Management System will use a simple, clean, and student-friendly design. The interface will make it easy for students to scan QR codes and for student officials to create events and monitor attendance records.

Primary Colors:

The application will use the official school colors of maroon and yellow.

App Logo Concept:

The logo may combine a QR code, checkmark, and student identification card. These elements represent scanning, attendance verification, and student participation.

Typography:

The application will use a clean sans-serif font, such as Roboto or Inter, to maintain readability on mobile devices.

General Layout:

The screens will use clearly labeled buttons, cards, icons, and organized content sections. Important actions, such as generating or scanning a QR code, will be prominently displayed on the home screen.

Navigation Style:

Suggested navigation for students:

- Home
- Scan QR
- Attendance History
- Profile

Suggested navigation for student officials:

- Dashboard
- Events
- Attendance Records
- Profile

Expected Output

At the end of the project, the team expects to produce a functional React Native mobile application that allows students to record and view their attendance during school events using QR codes. The application will use SQLite to store event and attendance information and will support create, read, update, and delete operations. It will also provide QR code generation and scanning, automatic date and time recording, duplicate attendance prevention, and tools for authorized users to view, search, and manage attendance records.

Proposed Project Scope

The scope of the proposed Attendance Management System defines the features and functions that will be included in the application, as well as the features that are outside the project's coverage. The application will focus on managing and recording student attendance during school events through QR codes, while keeping the system within the intended requirements of the project.

Included:

- Student attendance recording for school events
- QR code generation and scanning for attendance
- Creation, editing, and deletion of school events
- Viewing and searching of attendance records
- Viewing of student attendance history
- Adding, updating, and deleting attendance records
- SQLite database for storing student, event, QR code, and attendance information
- CRUD (Create, Read, Update, Delete) operations
- Automatic recording of attendance date and time
- Duplicate attendance prevention
- Simple and student-friendly mobile interface
- Multi-screen navigation for students and student officers

Not Included:

- Real-time notifications
- Web-based administration system
- Cloud-based database or online data synchronization
- Advanced user authentication and account management
- Attendance management for activities outside the supported school events